Student Name: Unnathi Vithlani
CS - 7346
Date: 06/11/2025

AWS Lab | Identity and Access Management

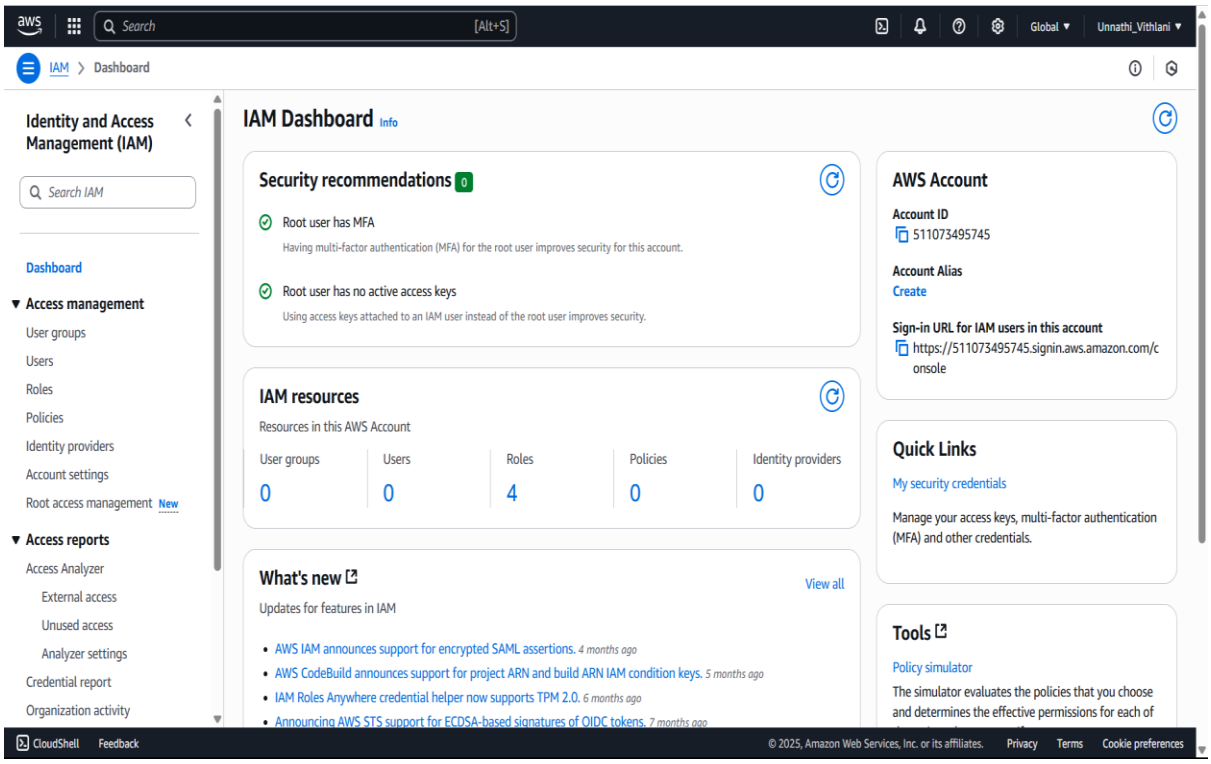
Lab Summary:

This lab is an opportunity to work with AWS Identity and Access Management (IAM). The lab will mainly focus on understanding how to manage AWS IAM Policies, to find the balance of security and access. The following are the specifics of the tasks completed in this lab.

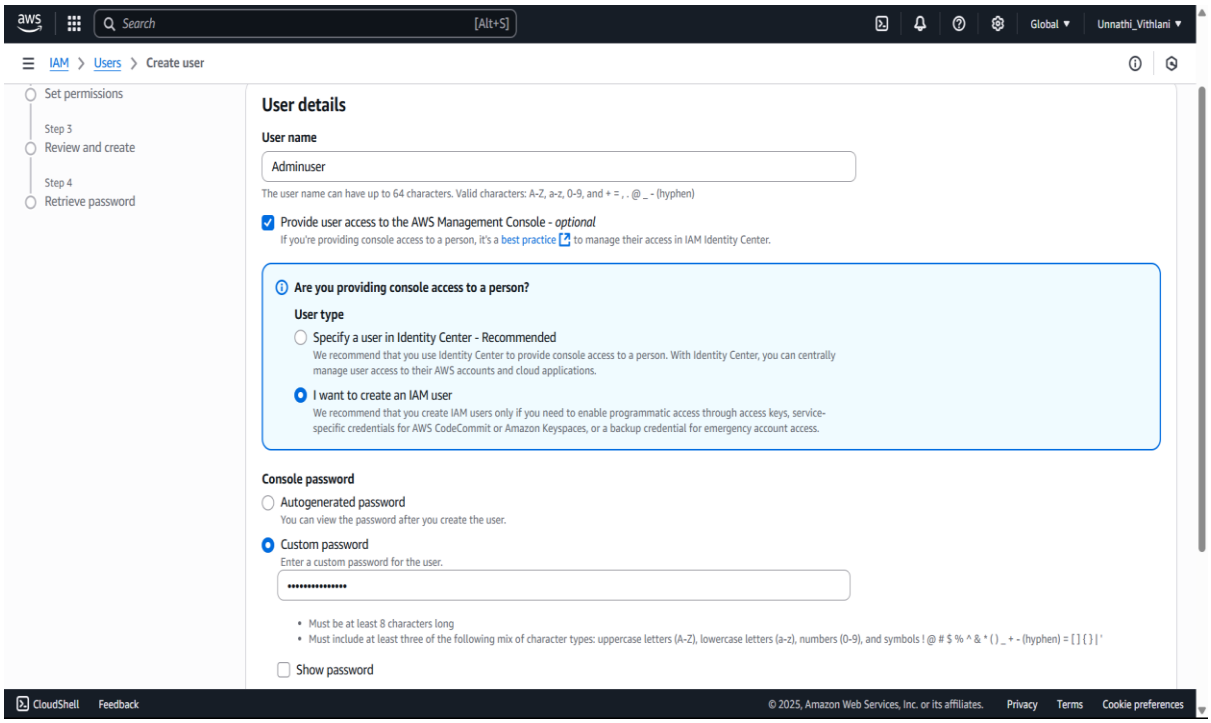
Exercise 6.1: Lock Down the Root User in AWS:

1. Create a Regular User and Assign the Administrator Access Policy

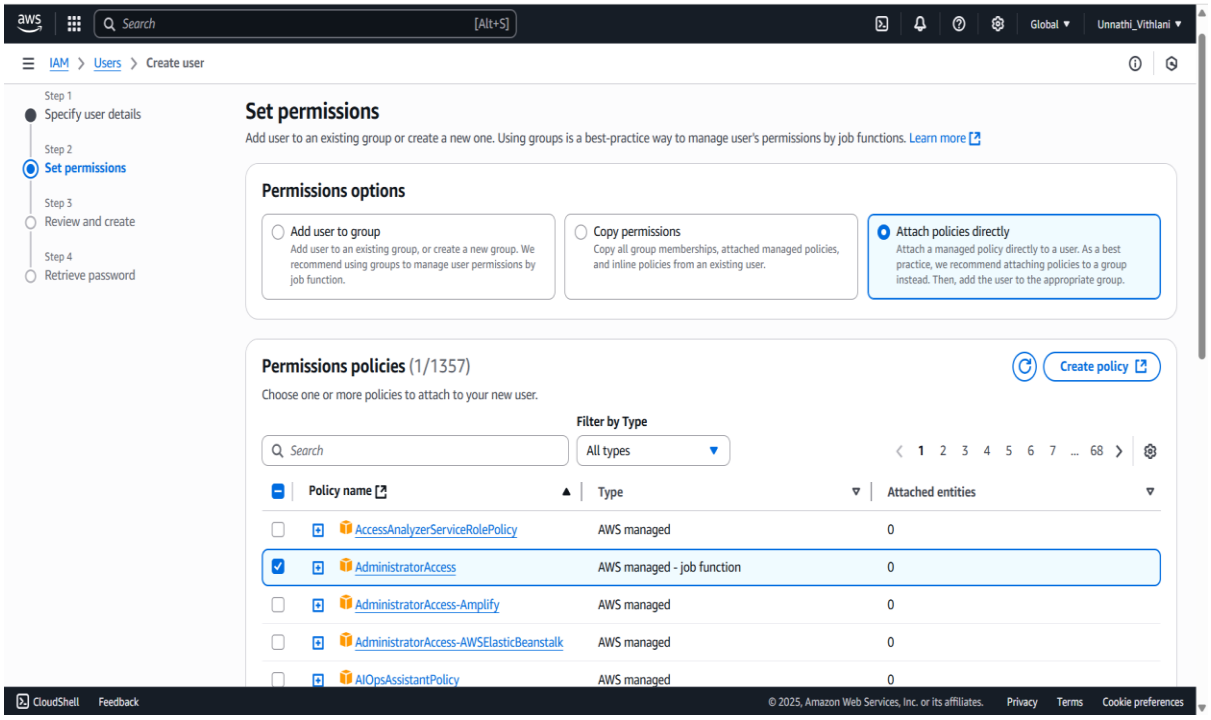
Login to AWS Console as the root user, from the AWS Management Console select IAM



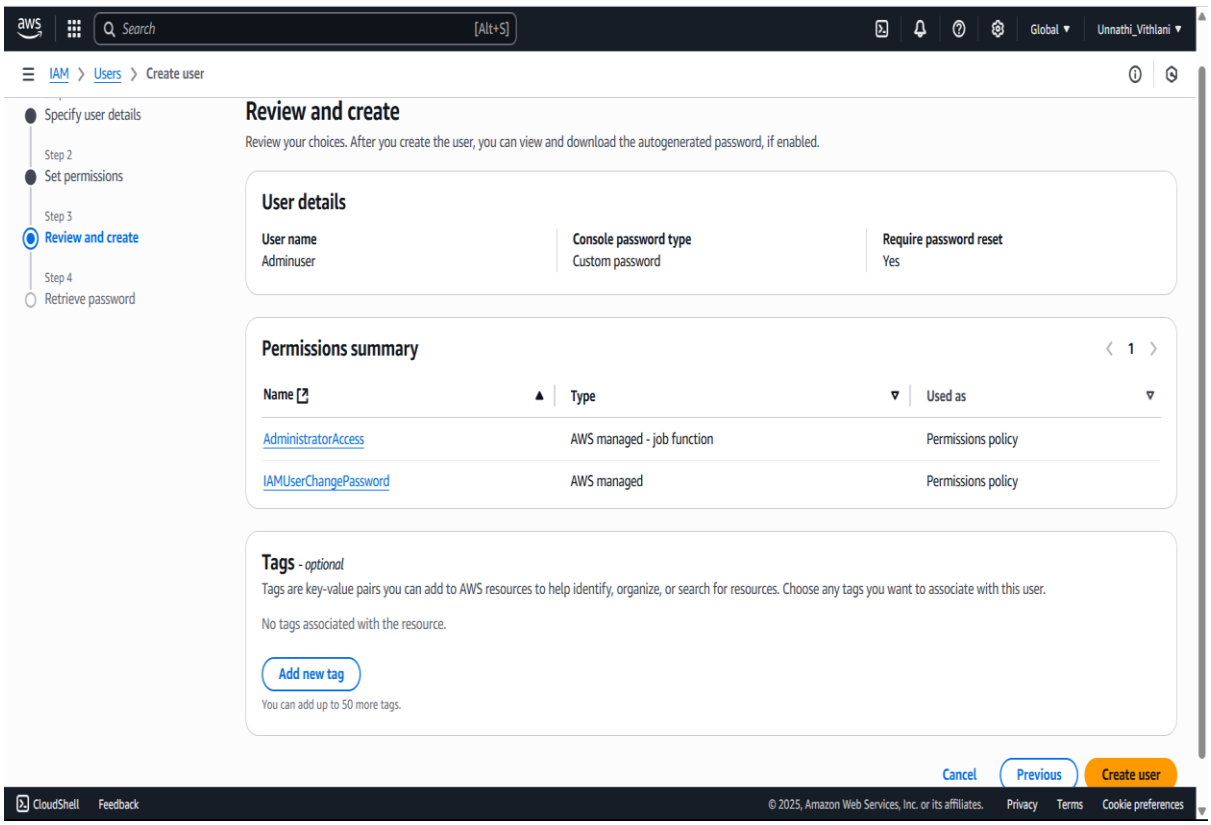
2. Click on the Users tab, then add users. Enter User name and choose custom password.



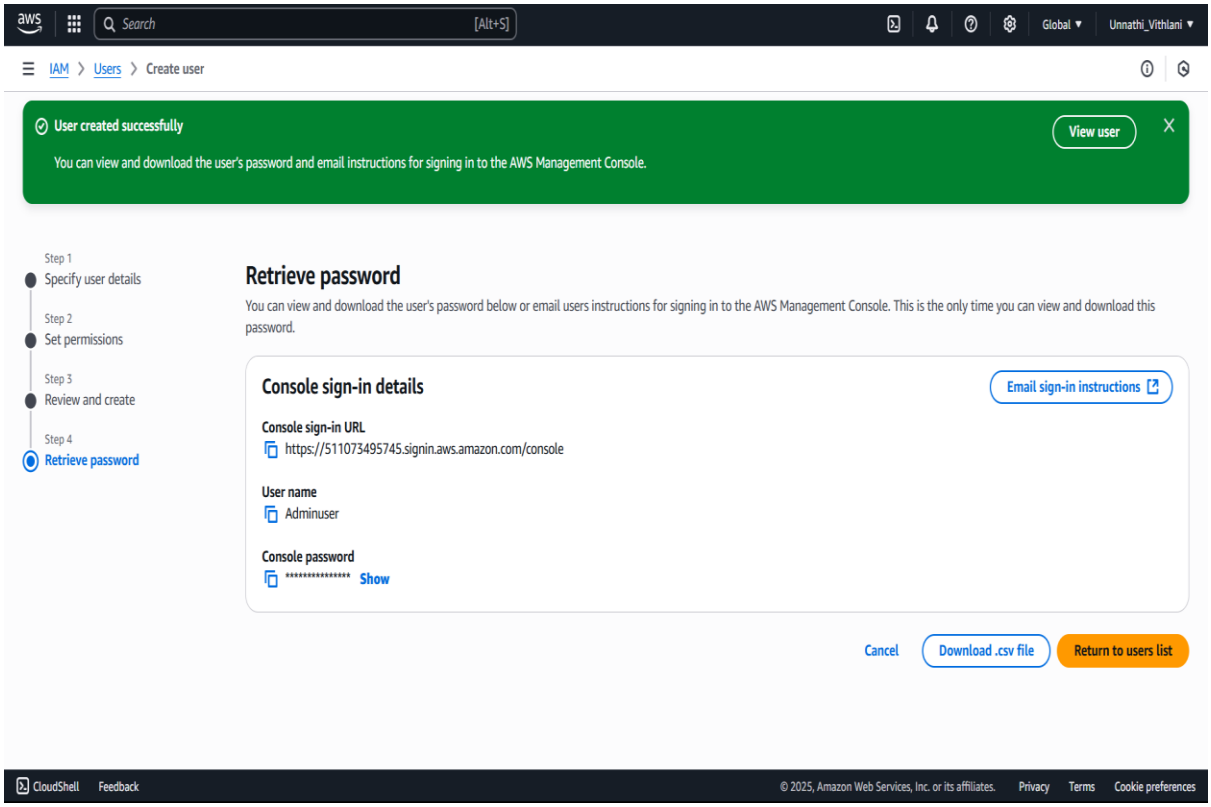
3. Choose Attach policies directly, check the box next to Administrator Access and select Next



4. Review question and select create User



5. Once created, Save login Url



6. Your root account should not have an existing access key assigned to it. Select Add MFA to enable MFA for the root account.

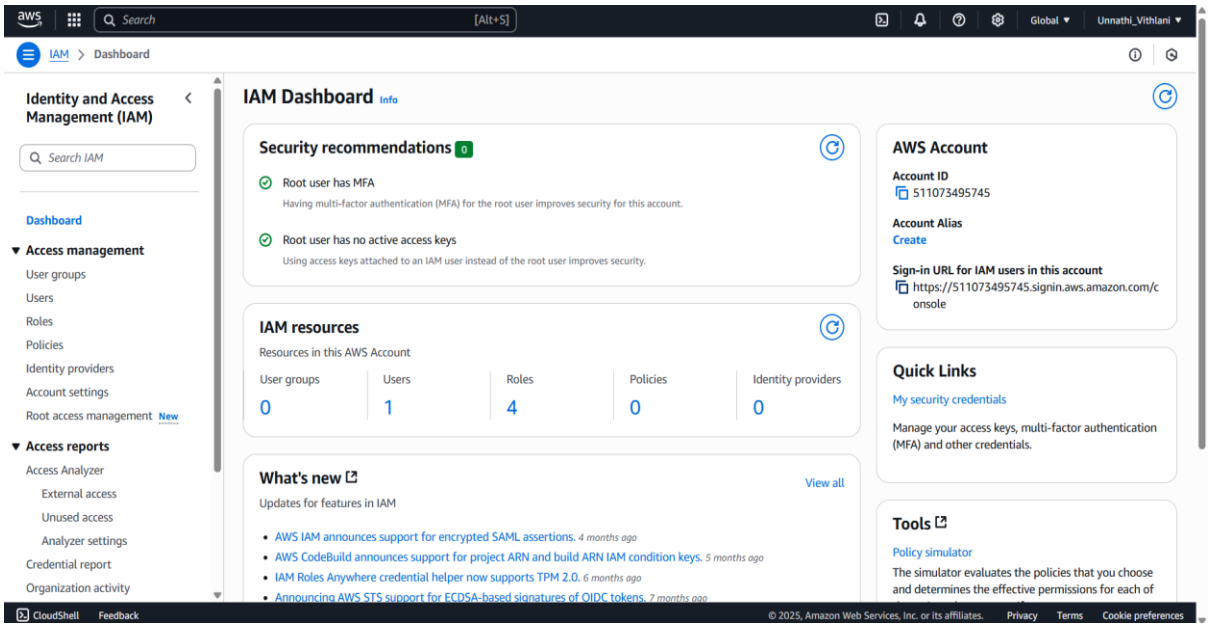
7. Then enter the name of the MFA device.

8. Choose the Authenticator app option. Download and install any authenticator app and scan the QR code on the screen.

9. Enter the MFA codes in your authenticator app.

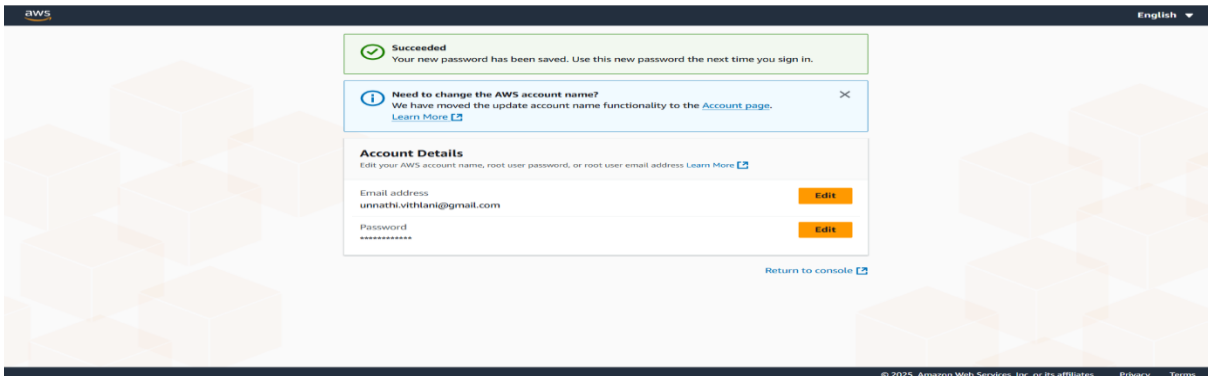
10. Click Add MFA, and MFA has been successfully established to your AWS root account.

11. Go to the IAM Console and sign in as the root user.



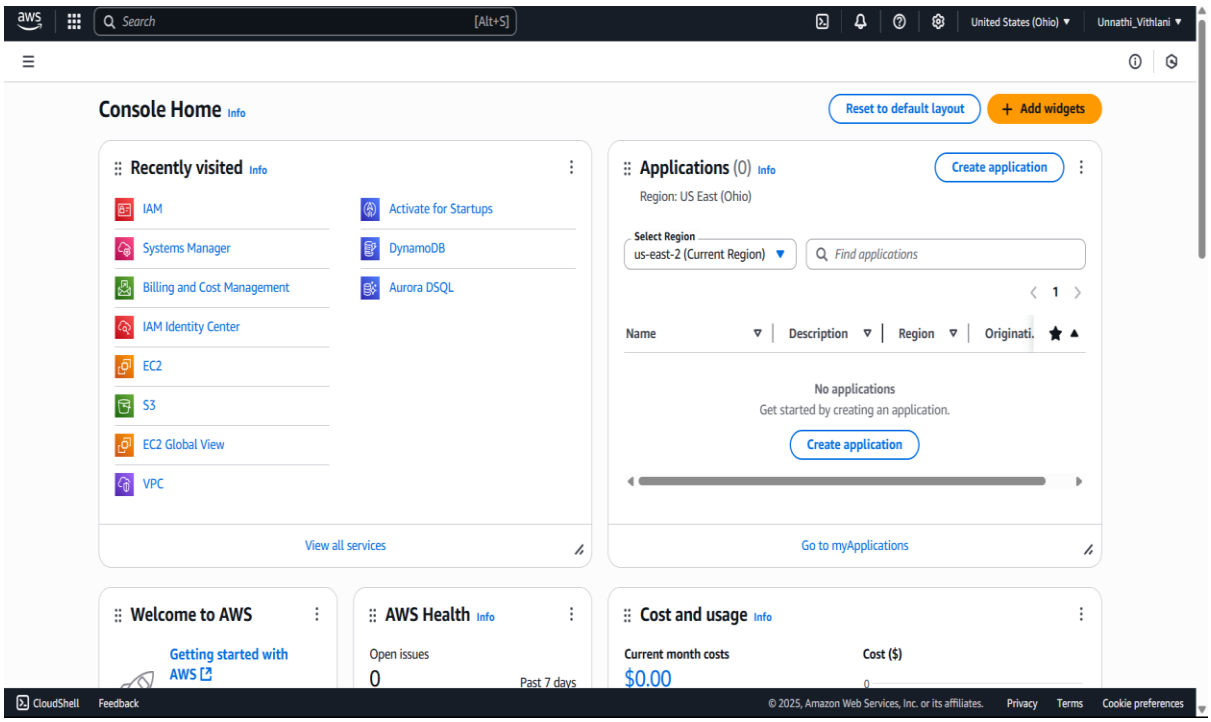
12. Change the root login to a password that contains non-alphanumerical characters.

13. Verify that you can still access the system as root, and then securely save Password.

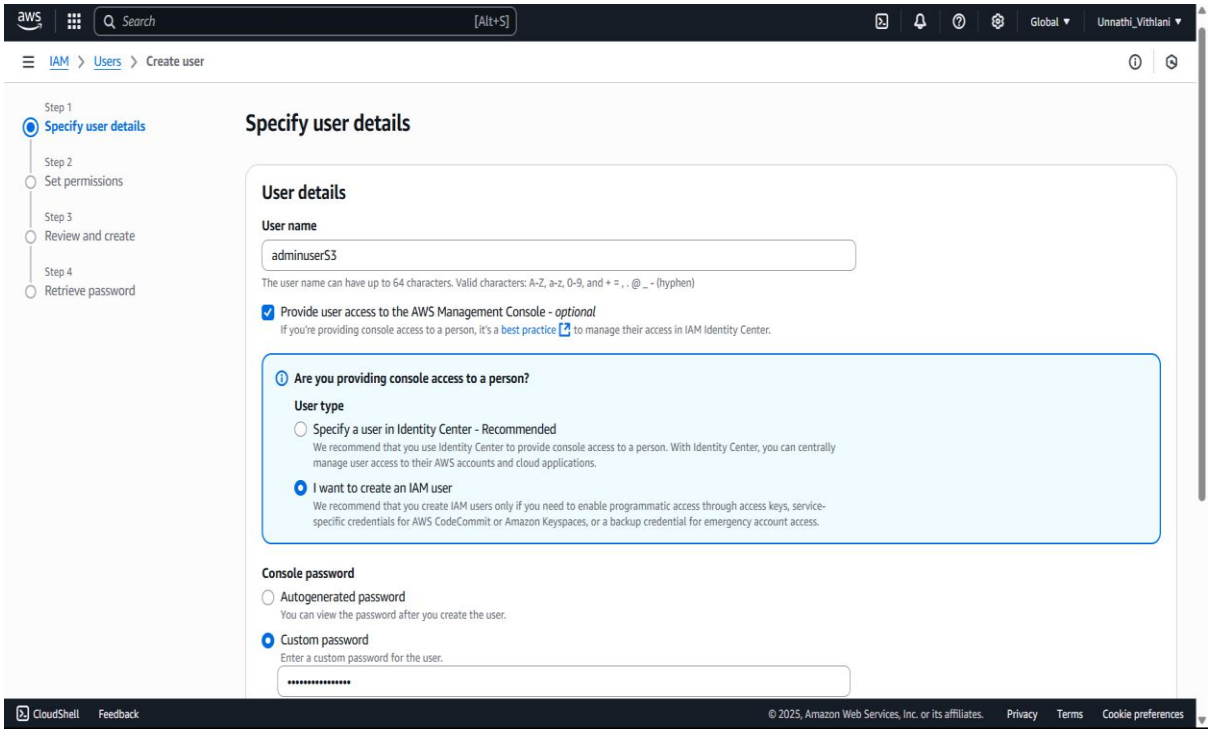


EXERCISE 6.2: Assign and Apply an IAM Policy

14. In the IAM Dashboard, you can create a new user by choosing IAM.

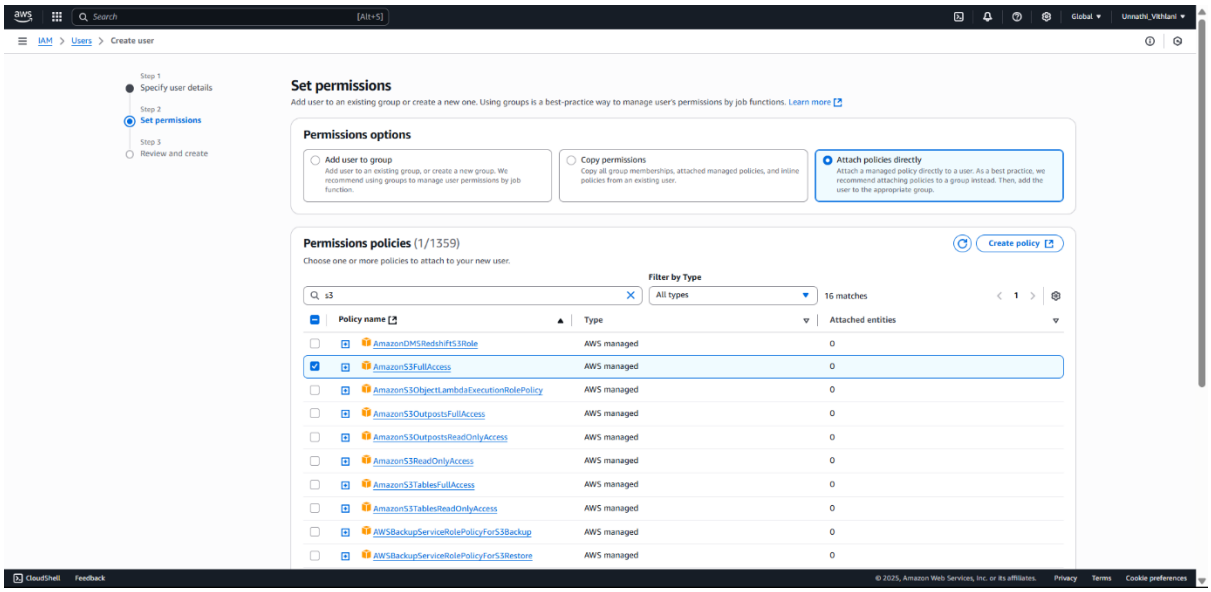


15. By entering the username example, "adminuserS3" you can select "I want to create an IAM user". You also have the option to create a unique password.

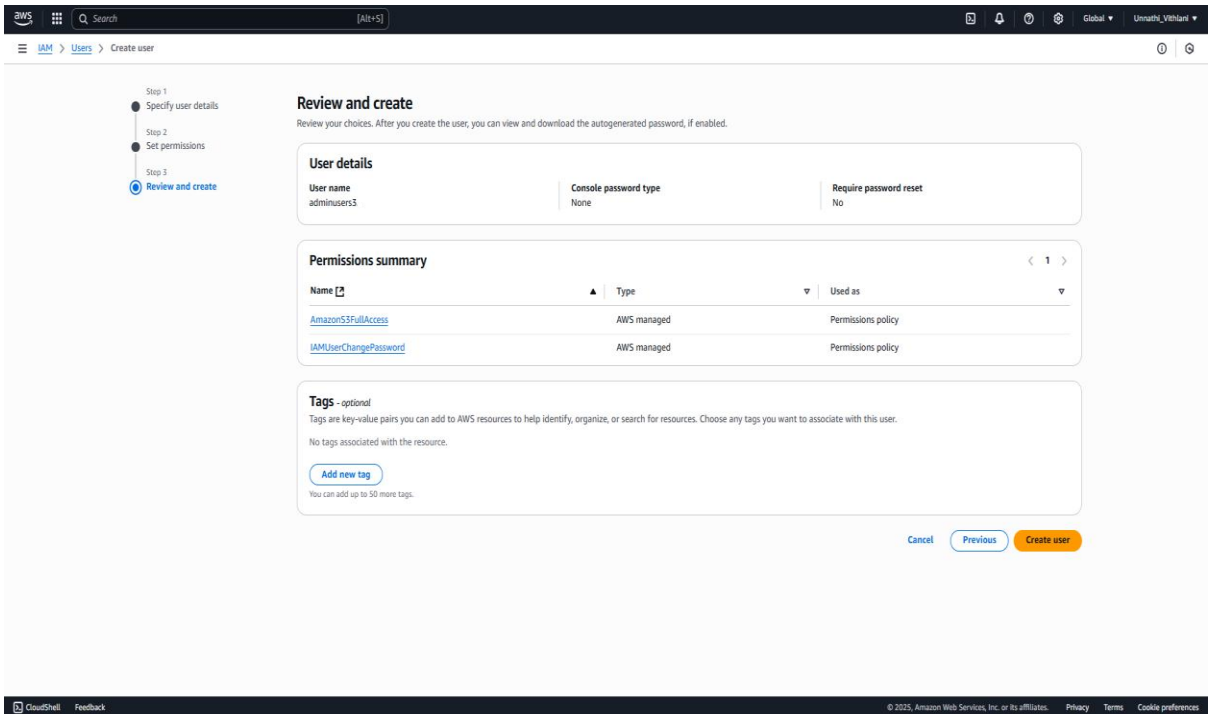


16. You need to assign the AmazonS3FullAccess policy, which allows users to add, change and delete S3 buckets.

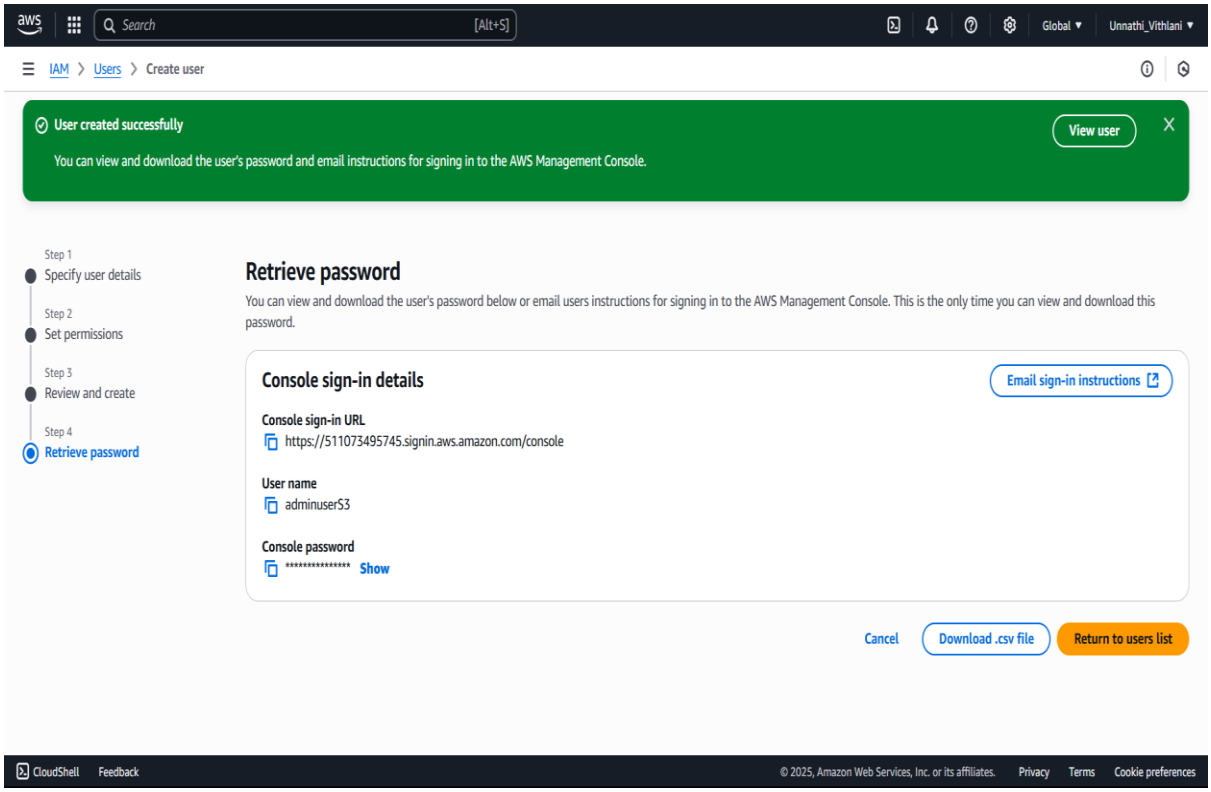
17. Select the "Attach policies directly" permission. The "AmazonS3FullAccess" permission policy was selected.



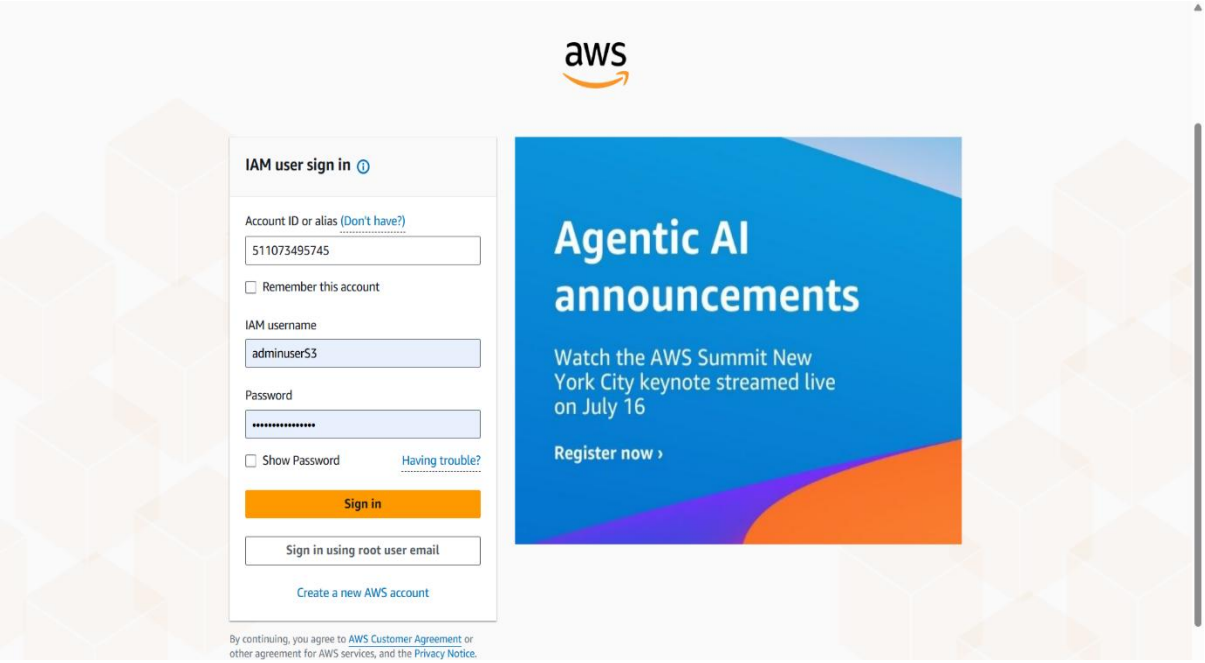
18. After reviewing all the user's information, select "Create" user.



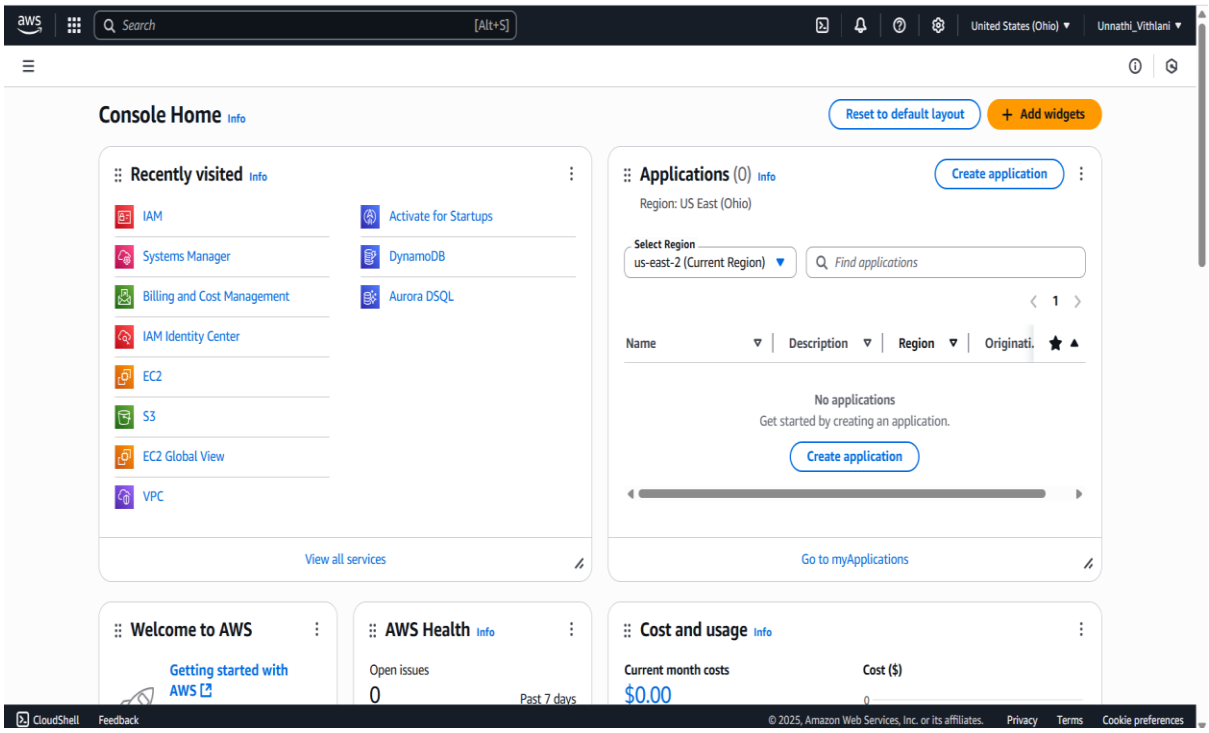
19. Take note of the directions for user login that will display.



20. After logging in, access the navigation menu and try to create a new S3 bucket as a new user.

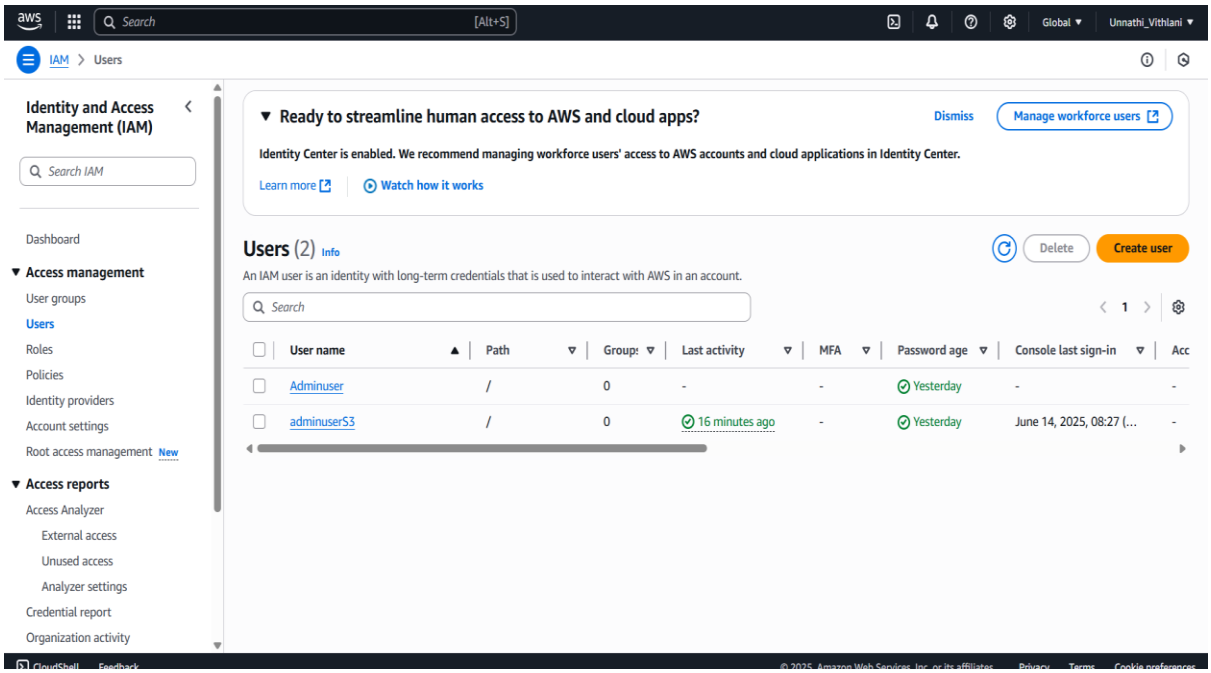


21. Now try to start an EC2 instance just to confirm everything is working. Your request ought to be turned down.

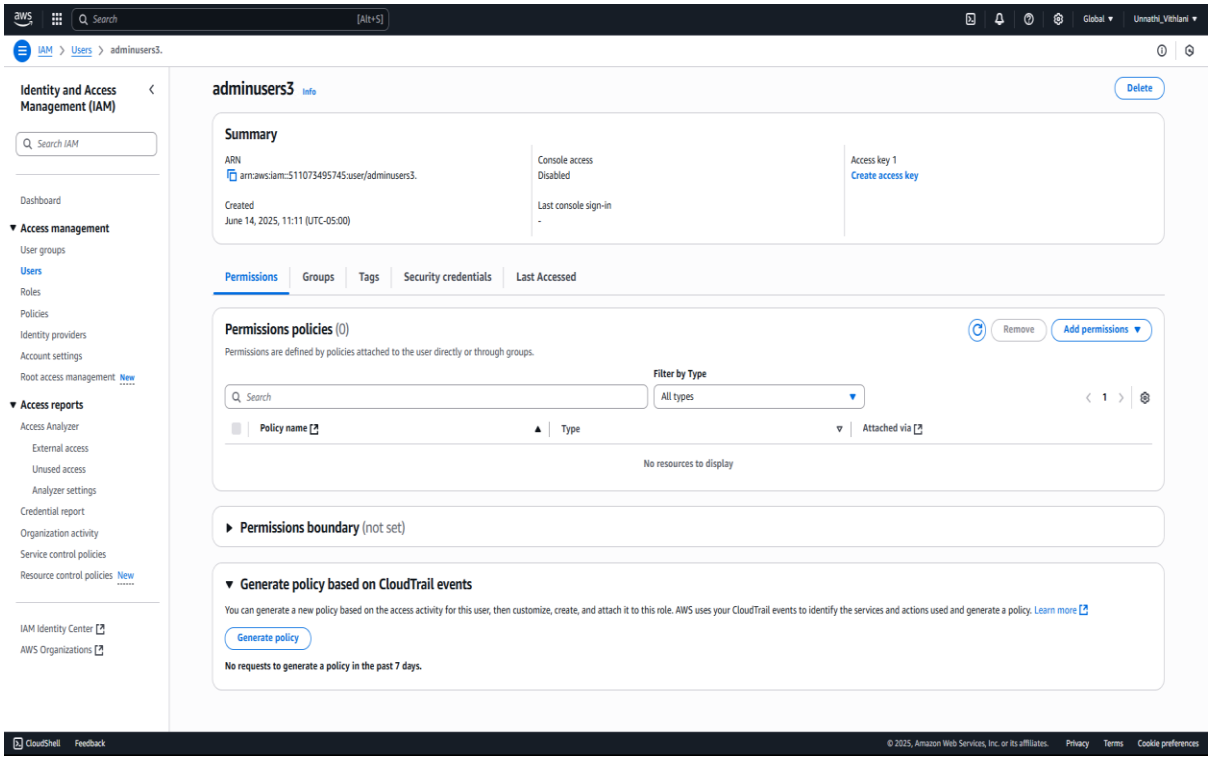


Exercise 6.3: Create, Use and Delete an AWS Access key

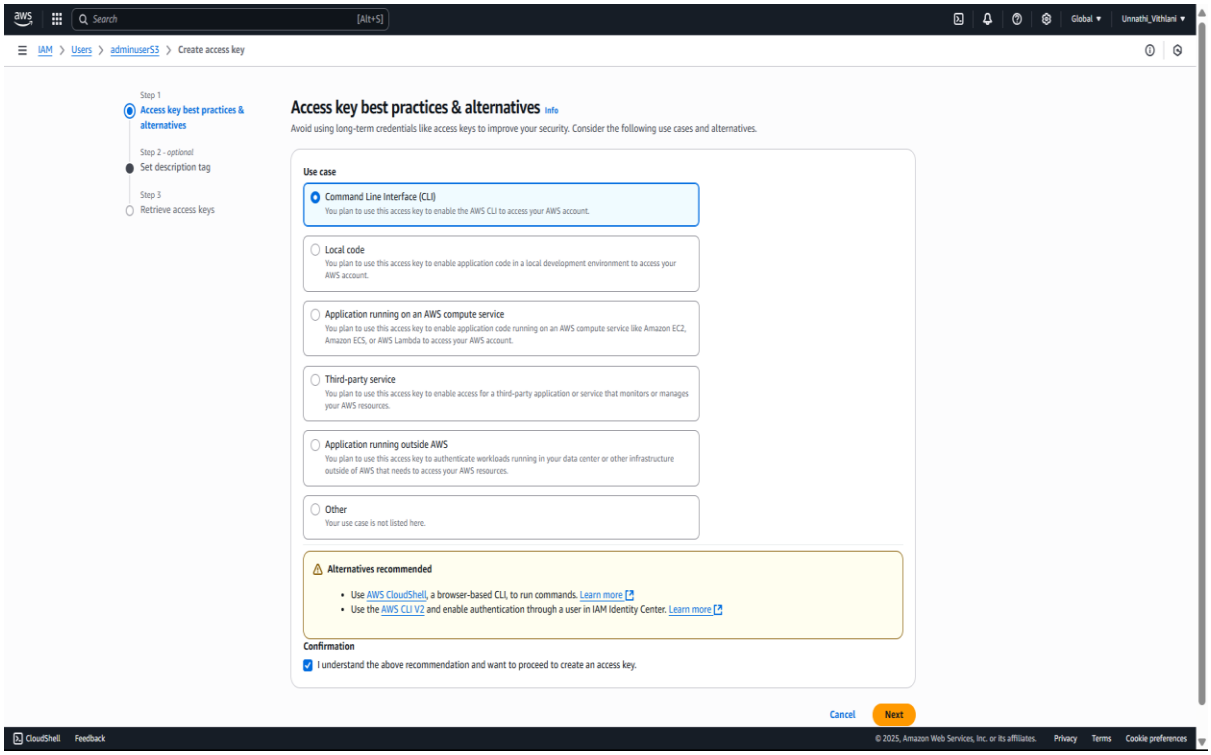
22. Create a new AWS access key, and write your secret access key and access key ID down somewhere safe.



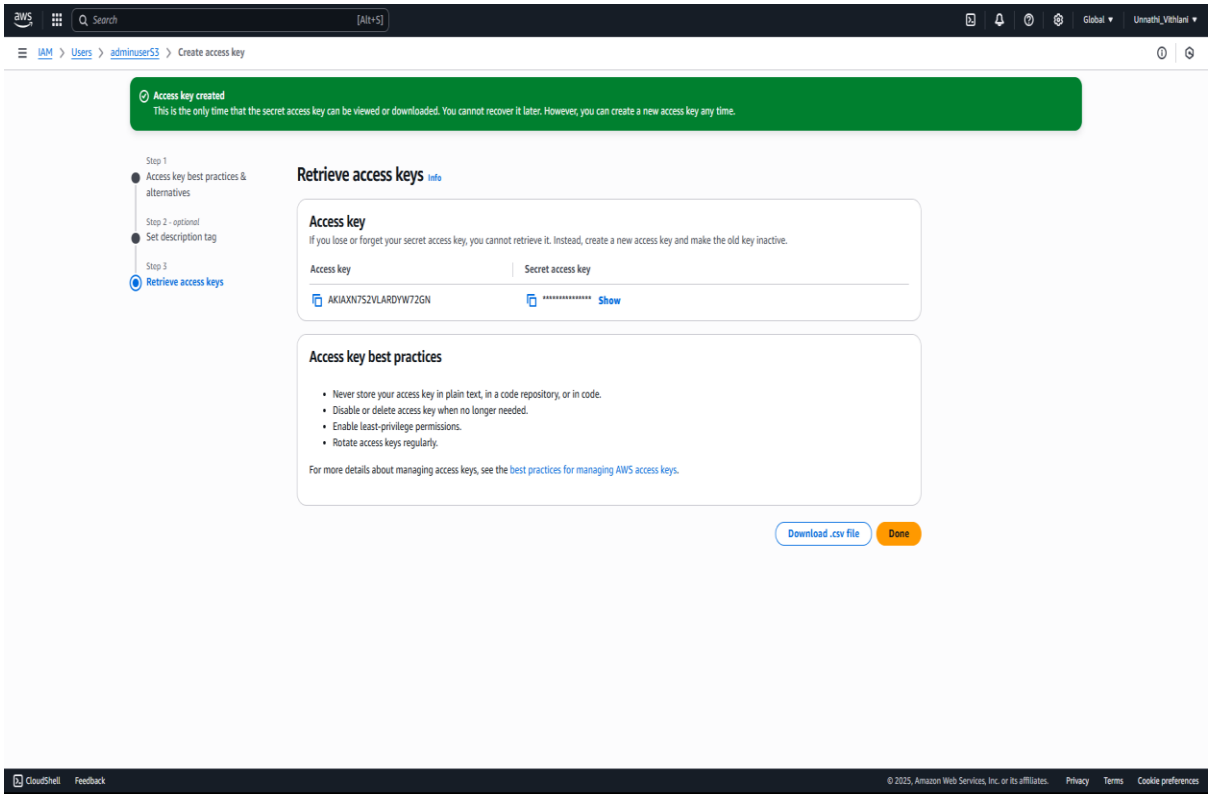
23. Go to Select Security Credentials and Create Access Key to create an access key.



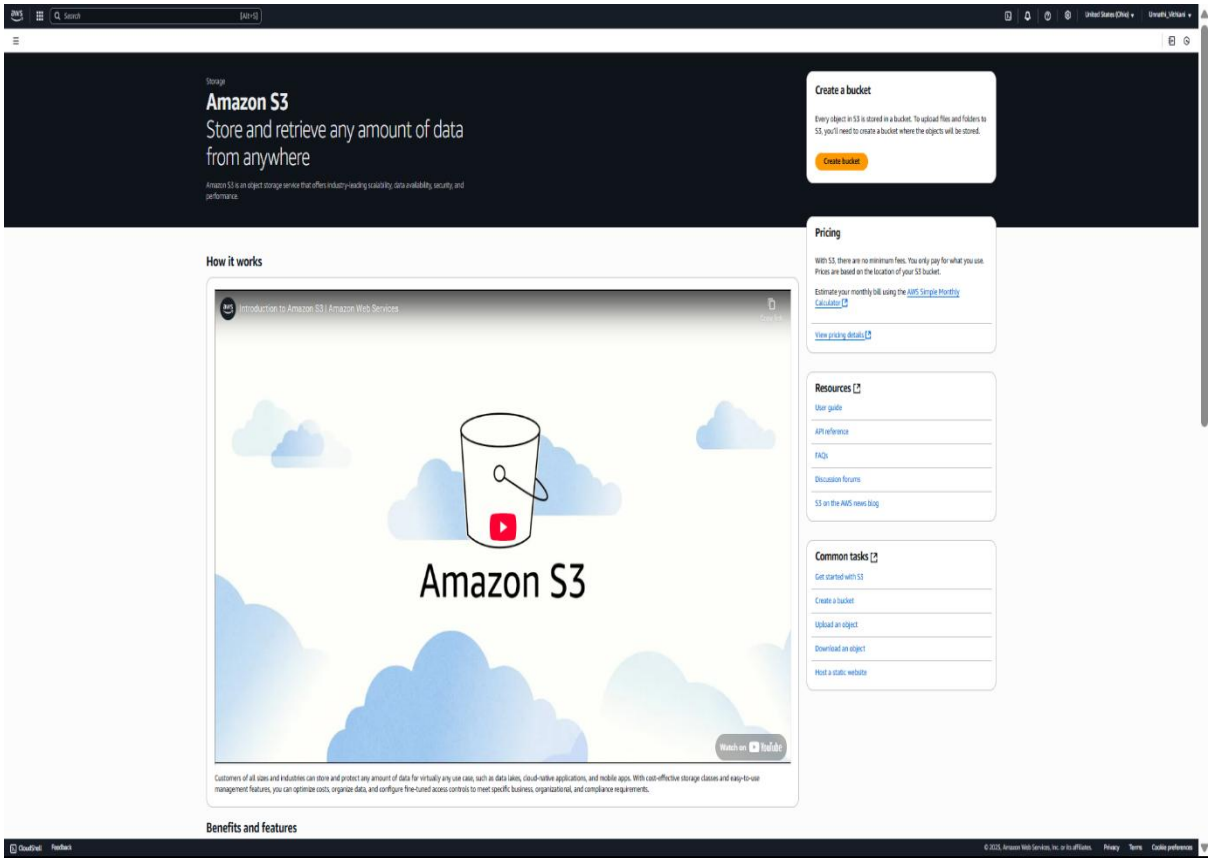
24. Select "Command Line Interface(CLI)" and then" Next." Use aws configure from your local command line to add the key to your AWS CLI config.



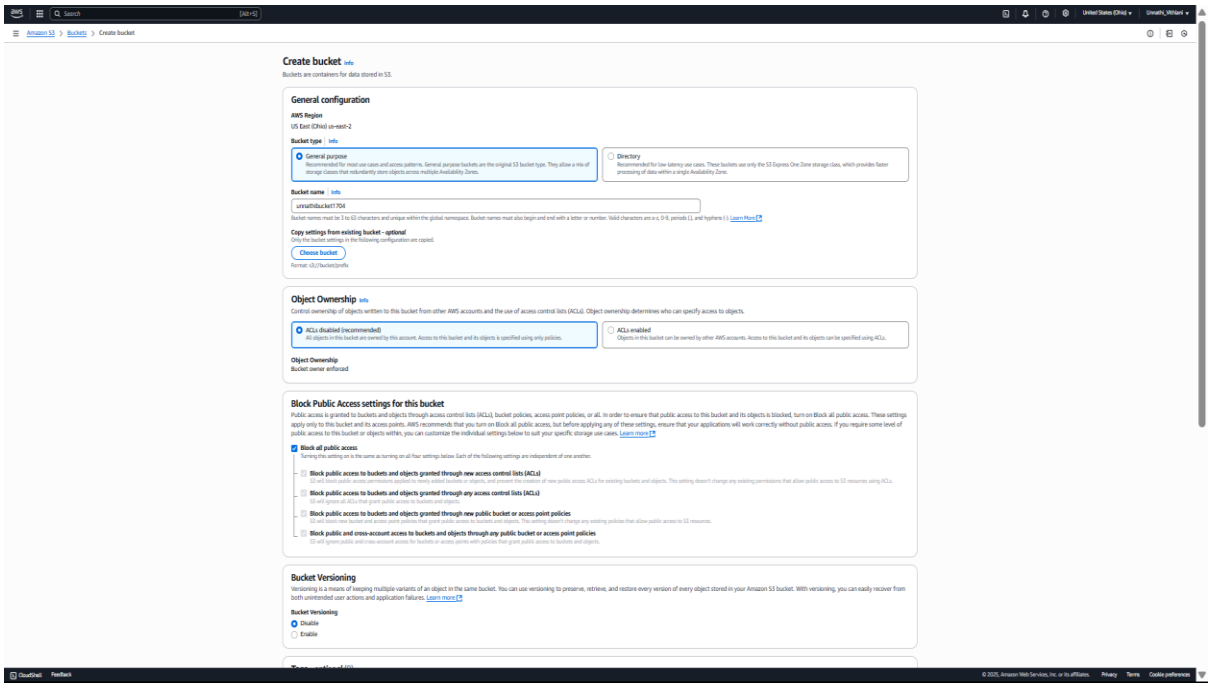
25. The access key is now created.



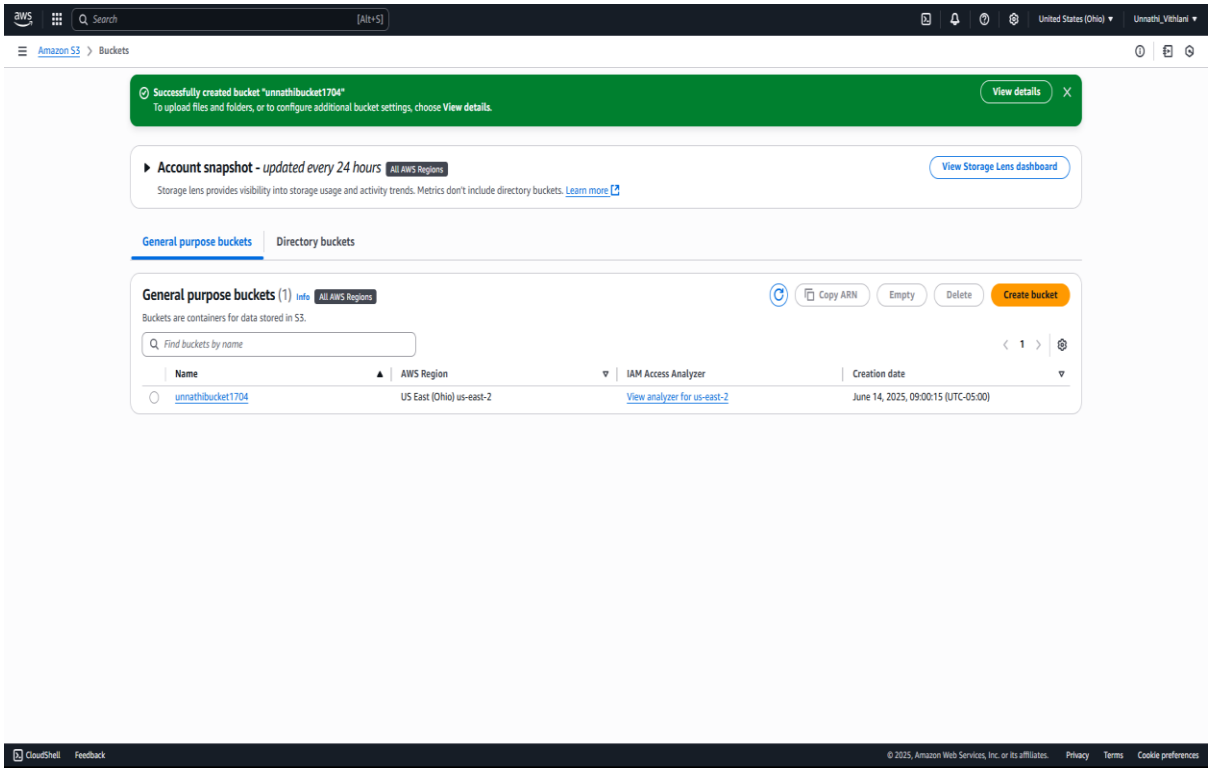
26. Next, try to perform a task, like view your S3 buckets, then use your new key and AWS CLI to upload a local file.



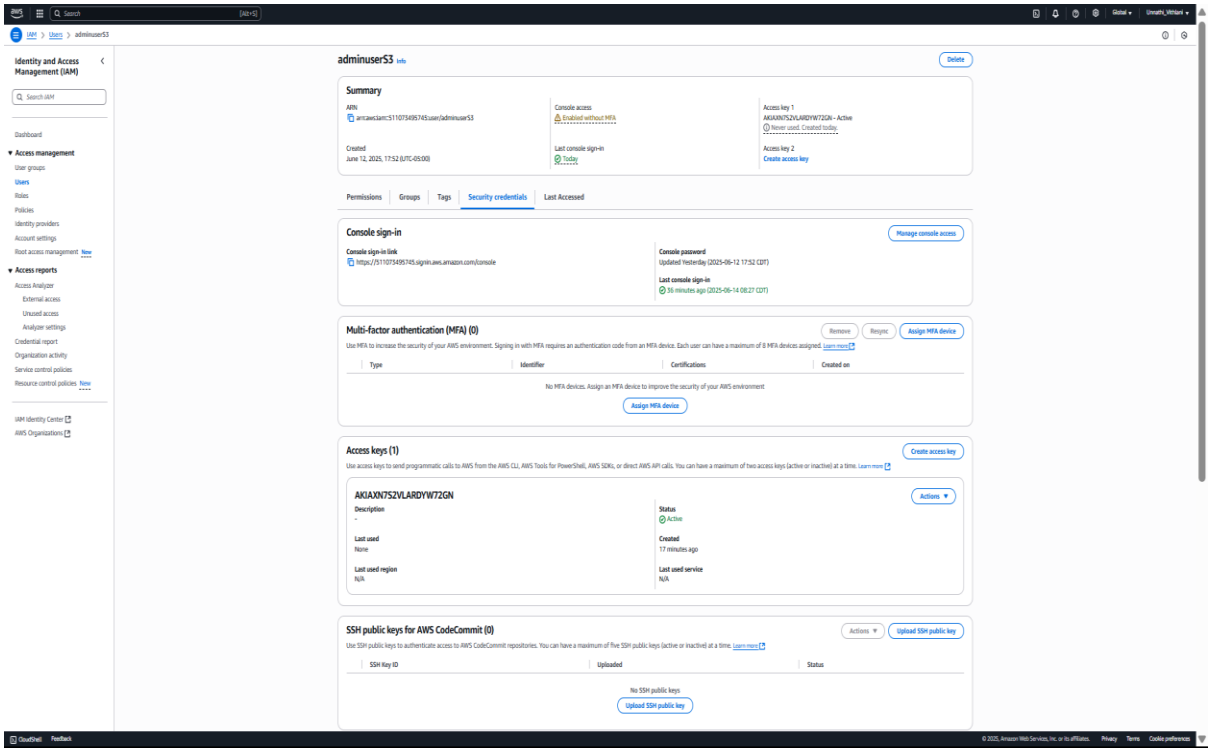
27. Select "Create bucket" to create a bucket, then create the bucket and give it a name.



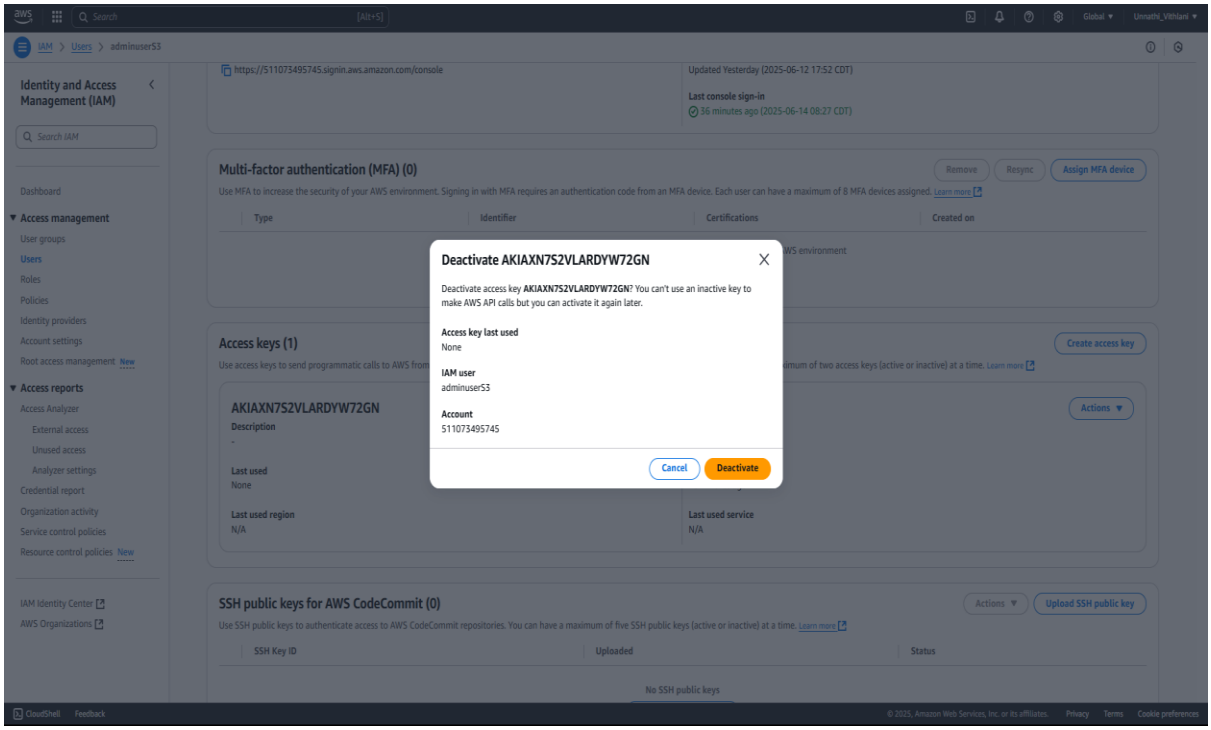
28. A bucket has been successfully created.



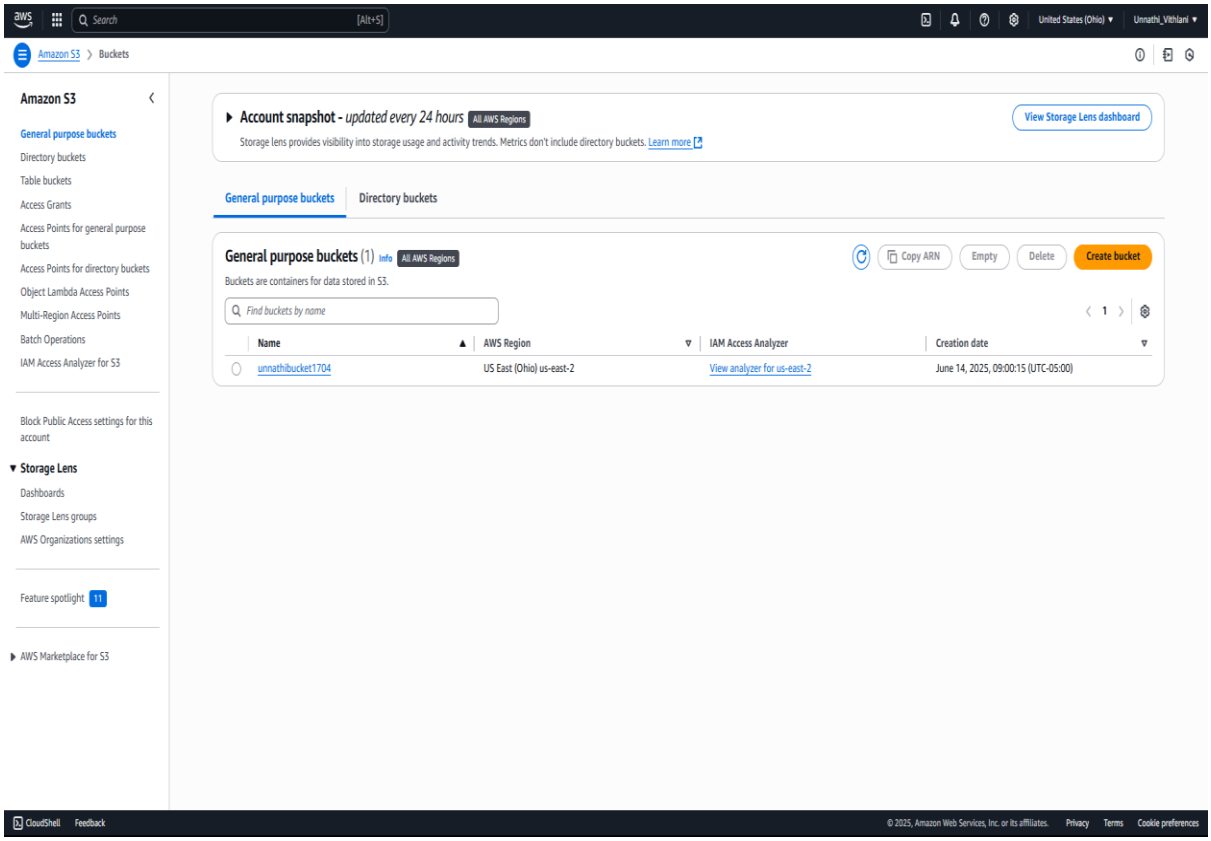
29. Go to your IAM Dashboard in the console and deactivate or delete the Access key you just created.



30. Confirm it is the right user name, then select "Deactivate."

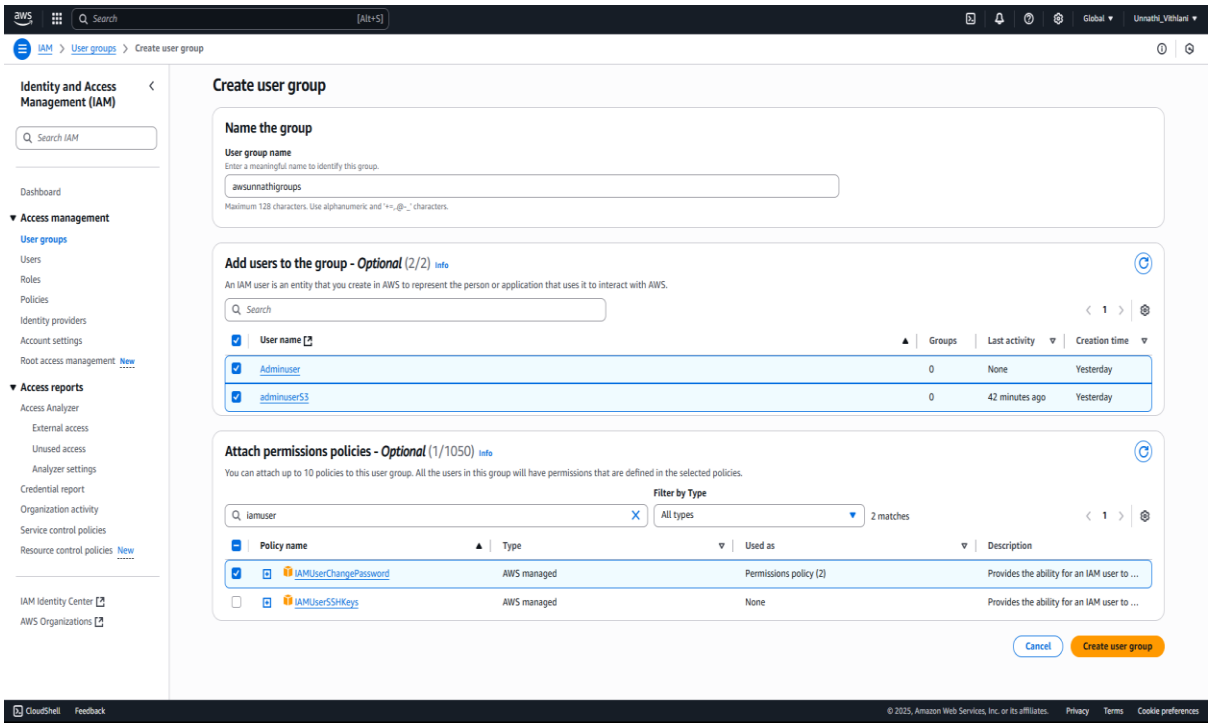


31. Validate that you can no longer utilize this key to do manage your S3 buckets.

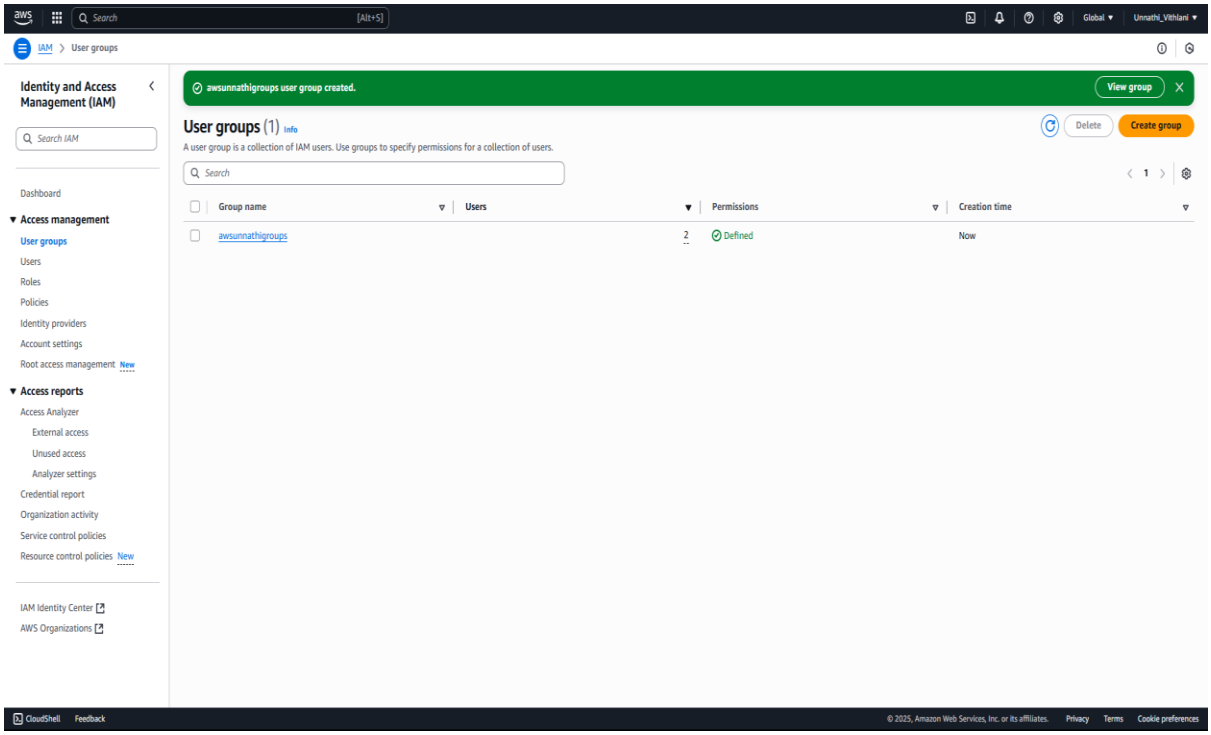


EXERCISE 6.4: Create and Configure an IAM Group

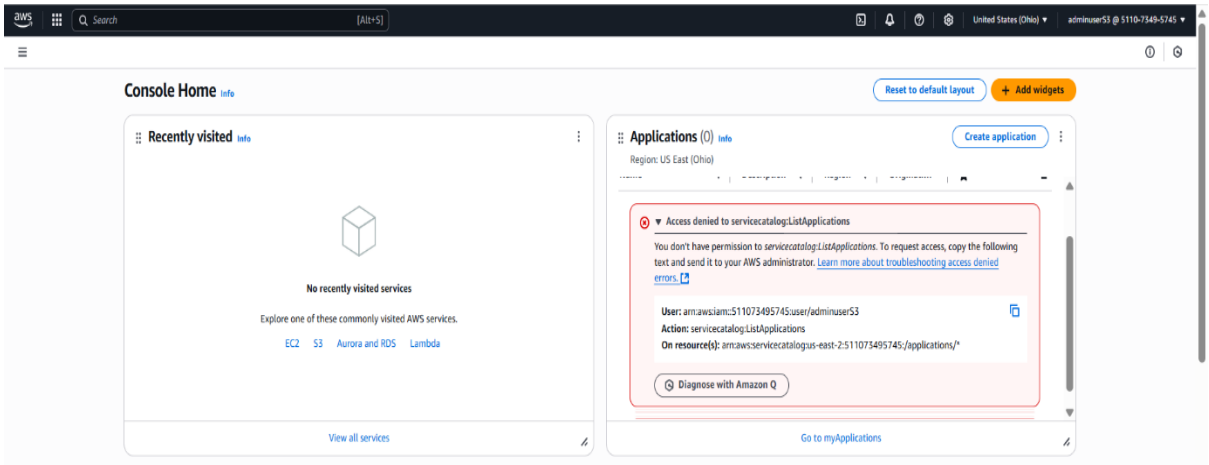
32. Make sure your account has at least two IAM users (including your two users in the group).

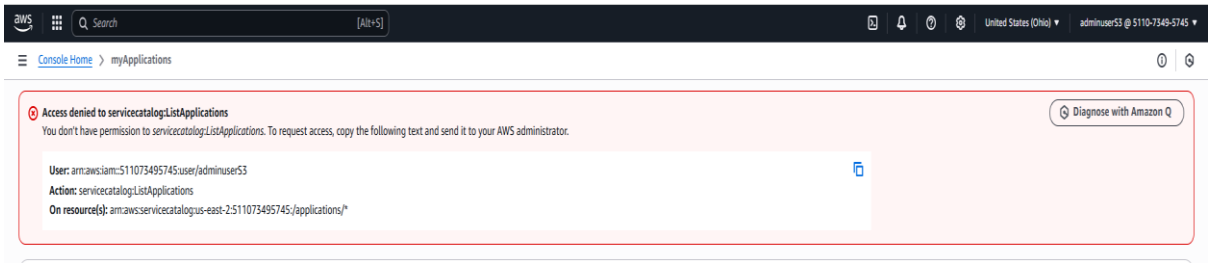


33. Create a new IAM Group, and it must have at least one policy, perhaps IAMUserChangePassword.

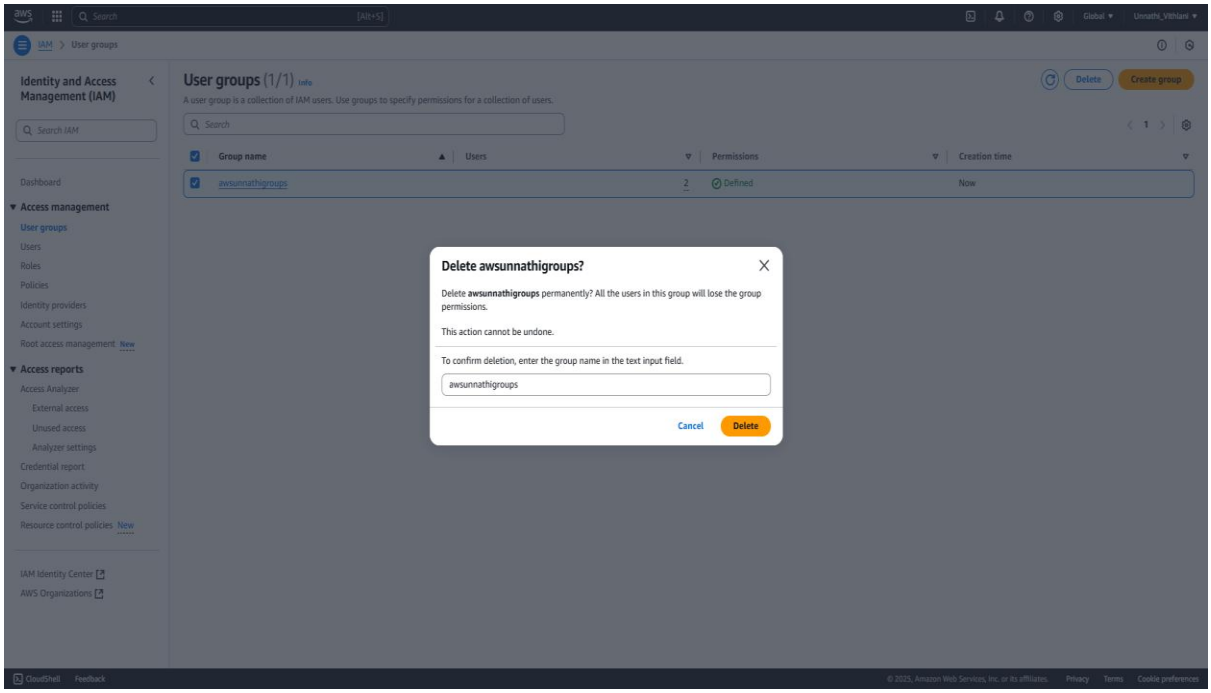


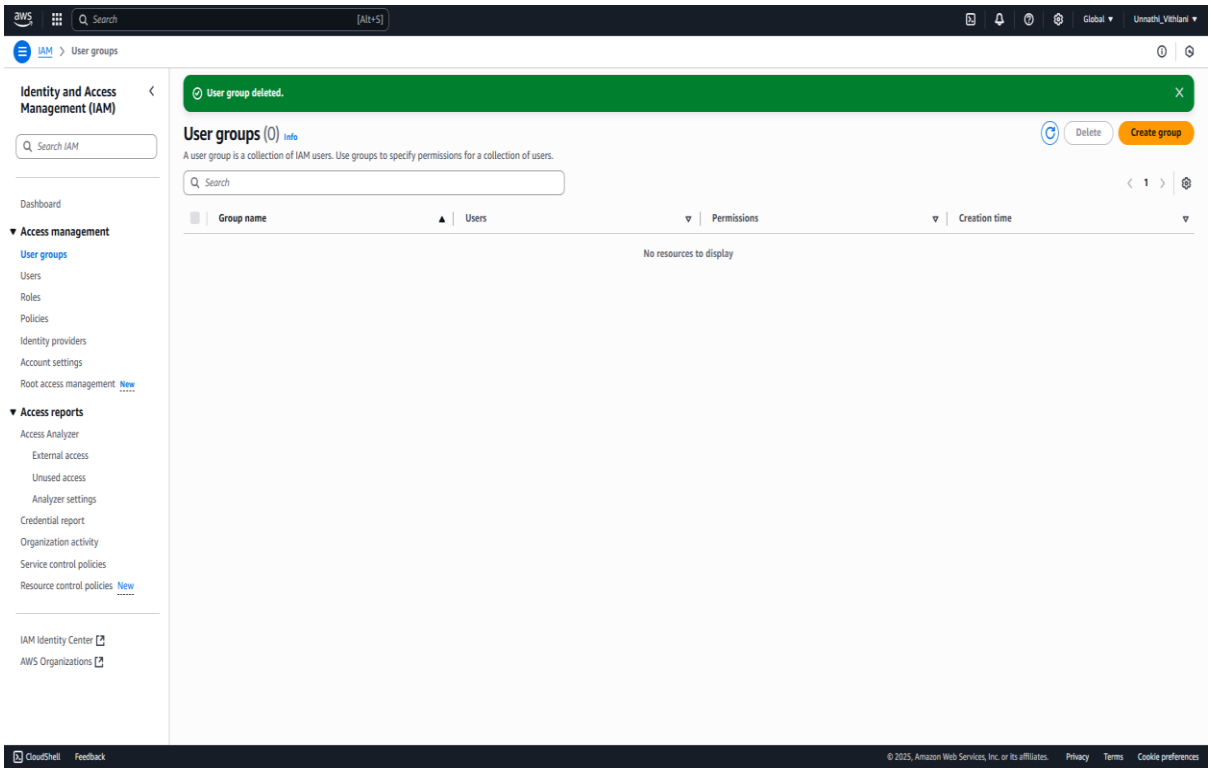
34. Access Denied.

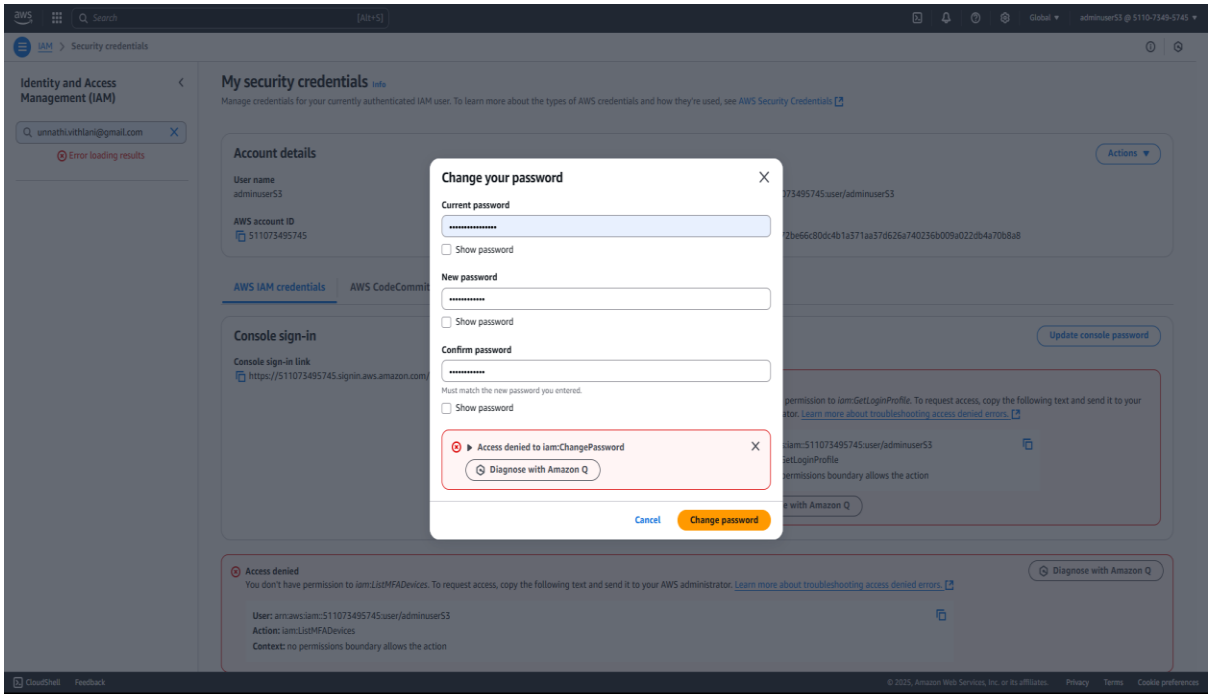




35. Now verify users can change their own passwords.





36. Ensure the users can no longer change their own passwords after you delete the group or changed the policies.

37. User group is effectively removed.